

NOTES FROM

Running to the Top

Arthur Lydiard

Many of the medal winners in the 1992 Olympics were not the best athletes but they were the best-prepared

for pre-game meals, you should have mainly carbohydrates, such as honey sandwiches, baked beans, etc., and be sure the meal has time to digest by eating at least three hours before you compete.

we have to develop oxygen uptake to make our body machinery function better.

B12 (cobalamin) helps to run the switchboard of the central nervous system, relaying messages between the body and the brain. Liver is the best B12 source but any animal product will do

Under aerobic conditions, one molecule of glucose makes thirty-eight molecules of ATP; but, when you train anaerobically, one molecule will make only two. So it is nineteen times more economic to run aerobically.

A lot of runners run with their legs permanently bent. They never straighten into that proper driving line. They're losing power all the time

walking about the house on the sides of the feet – doing a little often – is good conditioning.

You cannot build a house without a solid foundation; you cannot build good physical condition without a sound aerobic foundation

B1 (thiamine) helps the body to turn carbohydrates into glucose, which fuels the brain and the muscles. Any athlete who carbo loads before a race should take 5 milligrams of thiamine a day, which will turn all the pasta into energy. Green leafy vegetables, whole grains, beans, nuts and seeds are thiamine-rich and also contain plenty of the other B vitamins, except B12.

The belief that you are at a disadvantage when you race against altitude-trained people is a psychological misconception, not a physiological one.

the recovery process. The heat prevents stiffness.

the muscle, after one training stimulus during any one day, is unresponsive to any further stimulus on the same day

They are sincere. They are intelligent;

the week you are not going to race, you train harder, and the week you are racing, you back off

Very few runners, for example, can maintain a good knee lift throughout a race because they lack the muscular endurance which comes from well-developed capillary beds in the upper leg muscles. Once the knees go down, stride length shortens and leg speed dwindles

many coaches give team talks moments before a game begins, when players are hyped up and edgy and their minds are already out on the field. The best thing to do is to leave them alone

Demand sincerity. I have always believed in this. It doesn't matter how good athletes are, they can have the best potential in the world but if they are not going to be sincere about training and sincere to their coach it is better that they should be told to go and take a jump at themselves.

This exercise, slightly anaerobic, involves hill training, up and down the hills, using three different exercises – bounding, springing and running – in a co-ordinated routine, and you'll employ some timed running over the hills as well. The purpose in all this is to use your body weight against gravity to build more powerful, springlike muscles; and to use the downhill running sections to build leg speed by striding out fast. At the bottom, you throw in some wind sprints to begin the development of your anaerobic capacity.

No one can train hard and compete well at the same time; no one can fail to train and expect to play well; but everyone who does the basic preparation honestly in the off-season goes into the season at a peak of physical fitness and can then reduce training to hold that peak – and play right up to the limit of his or her ability.

em. If they have nothing to do, they'll probably get into trouble; if someone can give them a goal like going running or a game of football or cricket, they won't

fifty per cent of those hospitalised for heatstroke after intense exercise were potassium depleted. Many had taken salt tablets, which forced the potassium out of their bodies

Sprint the straights and float the bends.

But, to be successful at the highest potential level as soon as possible, every athlete in any sport needs a coach

The intake of up to 200 grams or half a pound of glucose or fructose during the two days before you play is sufficient. No more.

Not only did they improve as athletes, they also became better students. They were able to study longer and more effectively without suffering from mental fatigue

Vitamin C

The best carbohydrates are in starchy foods, grains and so on but they can cause wind and, if the carbohydrate need is high, you could not eat eight potatoes at a sitting the night before a race without suffering for it.

Try to make yourself six inches taller when you're running

Bring the laces from front to back without crossing them diagonally across the foot; they must cross over the top from the hole on one side to the hole directly opposite on the other side.

the number of mitochondria per cell may range from a few to more than a thousand

I have never been a fan or an advocate of supplements but, unless you can be certain that the food you eat contains everything your body needs, I have to concede that it is wise to take multi-vitamin and mineral supplements

This is one of the reasons why, if you're a jogger for the sole purpose of keeping yourself fit, it's much better to do it first thing in the morning, even if it means an early start to the day, because it gives a psychological boost to the rest of the day. You go off thinking, "Good, I've done my exercise for the day and I feel better for it." There is nothing wrong with jogging in the evening, except that, if it's been a hard day, you have to drag the effort out of yourself and it becomes a temptation to start missing days. The psychological bonus is not there.

Whatever sport you're involved in, you've got to adhere to and understand the mechanics. I recommend here a very good book by Geoffrey Dyson, 'The Mechanics of Athletics', which explains the mechanics basically and simply.

An atrophied muscle group, when trained, may show a rate of increase in strength five to six times as great as when we begin with an untrained normal muscle in a condition of normal strength

aerobic metabolism produces innocent wastes, water and carbon dioxide; anaerobic metabolism produces lactic acid, which, as it accumulates, progressively prevents the muscles from contracting.

Practical coaching experience

In the week you are going to race, you do some short, sharp wind sprints, a little easy fartlek work, an under-distance trial, maybe some stride-outs and some jogging. You cut your long, steady run down.

If there is any secret to the success our patients have had in treating their depression with running, it is that they have tried to run each day in such a way that they would want to run again the next day.

many coaches and athletes are inclined to follow training methods that proved successful years before and are not keeping up with the evolution of sport

demand sincerity. As a coach, your time is the most valuable thing you can give anyone; if that time is being wasted, you would be much better at home with your wife and family or making money for yourself. You don't have to be down at a track with athletes who are cheating on you.

The heavy anaerobic period also lasts about four weeks

The warm-up has two functions: To increase the pulse rate and blood circulation; and to warm muscles and reduce their viscosity so they will function better. To achieve both these objectives, you have to run on the spot or jog at reasonable aerobic effort for about five minutes and your body should be kept warm. A track suit over the playing gear, for instance.

The tall, willowy build, what we might call the Swedish type, is physically best equipped for running

bananas, oranges, tomatoes, cabbage, celery, carrots, grapefruit, apples, beans

They are dedicated. They have natural ability which suits their particular event. They have coachability. They have determination. They are ambitious. They have pride in themselves. Put all those together in one athlete and you have a world beater.

the best coaches take longer to produce the best results.

Taking calcium gluconate before a game and during the training week is helpful because muscle contractions require calcium ions and, unless you provide enough, you can experience cramps and nervous reactions, such as edgy, jumpy muscles which play up at night and cost you valuable sleep.

if you can get an athlete in his teens to train but not race until he is mature, you will have laid the foundation for an Olympic champion

not frozen or bottled juices

The key to good running is relaxation. You must be nice and loose in the shoulders. The arms should be loose, relaxed and coming straight through as they do when you walk. The hands come from behind the hips and, when the elbow gets alongside the torso, the arms should flex so that the thumbs are in a line directly in front of the shoulder-blade.

Spend 15 minutes a day on stretching and suppling – I'm no fan for much more than that – but it does become an essential requirement to avoid torn muscles and tendons. It gives support to the tendons

Knee trouble arises during conditioning work when the upper front leg muscles and sinews are too tight and stressed. These quadriceps must be strengthened by uphill running or by doing squats.

A marathon runner, employing a moderate work-rate, can get enough oxygen to economically burn fat and glycogen. This enables ATP to be rebuilt as fast as it is being used and the trained runner, working aerobically, can continue for several hours – in the case of the elite ultra runner, for day after day of steady aerobic output.

ninety percent of the colourings used by food manufacturers were synthetic and most of them were coal-tar derivatives

people who take up jogging, whether it's just for jogging's sake or to help their golf or some other sporting activity, meet so many other people from all walks of life who have a common interest in just feeling good about themselves.

To run correctly at speed, we've got to get the knees up so that the thigh is as near horizontal as possible

Running for long periods and hill training give you that endurance in the legs but other muscle groups which also need this development must be tuned by weights training. So, during the conditioning period, you need two or three sessions of weights training every week, as shown shortly.

every metabolic reaction in the human body depends, directly or indirectly, on oxygen

The ability to motivate

It is best, after a half or full marathon, not to do anything fast for about two weeks. Just jog nice and easily right through that period

consider whether you really want to run. Some people just don't. Consider the options and find something you'll be happy with because enjoyment is one of the critical requirements of this project.

make coaches realise they had to motivate their players and appeal to their intelligence to keep conditioning out of season, during vacations, all the time, if they want to be real successes.

I have mentioned the importance of the respect and esteem of the athletes for their coach

you don't need to measure the distance you're covering or count the number of times you run it. Keep on doing them until you hit the wall;

This usually takes about 15 minutes and it is a most-essential safeguard against straining tendons and pulling muscles. It is not to be achieved in the first 15 minutes of the match you are playing.

Children can be started as runners as young as five or six, around the parks, jogging with their parents and so on and there's no reason why they can't go into short sprints at school or with track clubs. Sixty or seventy metres is the

Salt in excess is thought to contribute to high blood pressure. For athletes training in hot weather, this is bad news.

The coach should appeal to an athlete's intelligence, explain clearly what every training session means and what they are trying to achieve.

A smart athlete soon knows if he's being led along the wrong track. We all know our limitations, we know when we hit the wall. But only the smart ones learn how to handle that knowledge efficiently. If you go out to do a specific bunch of repetitions or a long hard run because that's what the coach has ordered, you'll know when the effort begins to get beyond you and you start thinking, "This is where I've got to quit."

During a game or just before it, any of the specially prepared drinks that can be bought from the pharmacies are better than merely sucking an orange. These drinks have been developed to replace salts and minerals lost through perspiration. They are extremely valuable on hot days when your perspiration level is extreme.

In the US, when youngsters show any form at high school, the tendency is to put them on the track and pile the anaerobic work into them. Consequently, they don't develop.

Vitamin E oil cleans up chapped lips, surgical incisions, burns (including sunburn), bedsores, warts, athlete's foot, poison ivy, any skin complaint.

the less we interfere with the natural movement of the feet, the better we are going to perform.

Gentle massage or applied heat, such as swimming or a sauna, alternating cold and hot, will help that suppling programme.

doing enough suppling and stretching exercises. The importance of the full extension of muscles and sinews must be understood

even when you begin a run slowly, you are likely to be running anaerobically until your aerobic mechanism reacts and takes char

fresh fruit and vegetables

We should all proceed like children in training, letting ourselves be guided by our own reactions. We should understand our limitations and keep within them. If we do that, particularly when we are training, we will get better results in the long run.

The less you bring your upper body into running, the better.

As a guide, it is wise to begin with weights you can comfortably raise ten times in succession.

Aerobic exercise requires the presence of oxygen; anaerobic exercise does not

It helps to be able to work out with athletes. You are young enough or fit enough to go with athletes during the conditioning phase and work with the slower ones, it is going to build their confidence in you. And while you're running with them, you are able to find out their potential more easily

the main requirement is back-bending but you need to hold the position for a minute or so to get real benefit. This, combined with sit-ups, will produce very strong back muscles in the right places to eliminate back problems.

Look at cardiac development as a progression which lasts for several years, even the rest of your life.

Rest for about a minute between sets and three minutes between exercises. Keep a record of your training, listing starting weights, number of reps and sets and be careful to note upward progression.

You do run better and you do feel better being close to nature like that. You train at an immeasurably higher rate and gain more benefit.

In conjunction with that anaerobic overload, you will work on your leg speed with typical sprint technique running and leg speed exercises.

Anaerobic training is what destroys young runners

The iron in meat also aids the absorption of iron from other foods.

Athletes need to enjoy their training

This comes back to listening to your body as well as your coach. Coaches cannot always know exactly what everyone should do; they may be fairly good judges but the amount of anaerobic work a runner can handle will vary from day to day and from athlete to athlete.

Morehouse and Miller ('Physiology of Exercise' authors), of the University of California

The Swedish runners had four to five capillaries per muscle cell in their quadriceps but the Kenyans had seven to eight

Calcium, vital in checking osteoporosis, which hobbles millions of elderly, particularly women, should probably be given as a supplement from about age 25, when bone loss begins

Look for a shoe that's flexible at the arch and has a good amount of resilient rubber underneath it to absorb the jarring. A shoe that feels comfortable with no pressure points. Forget the rest.

Triathletes don't have much trouble because their cycling muscles, swimming muscles and running muscles all balance each other perfectly

When it happens, you can put your finger accurately on the spot. There will be internal bleeding there and you must stop it. Ice or cold water is most effective. Treat it this way for three days before you begin massage.

oxygen your muscles can use is limited to your maximal aerobic capacity, or maximum oxygen consumption, known as VO₂.

Orange, yellow and green fruits and vegetables such as carrots, sweet potatoes, spinach, apricots and canteloupe

It is also important, if you're into a heavy anaerobic section of your training, to give yourself a minimum of 48 hours, or more if possible, for recovery.

Relaxation is the key to good running.

Two-arm barbell military press. Approximate starting poundage 50, 55, 60 lb (23, 25, 27 kg). Three sets, 3 to 5 reps. Concentrate on fast explosive-type presses rather than slow deliberate movements. Biceps curl. Approx. starting weights 40, 45, 50 lb (18, 20, 23 kg). Three sets of 10 reps. Full squat with barbell on shoulders. Approximate starting weights 70, 80, 90 lbs (32, 36 41 kg). Three sets of 10-12 reps. Concentrate on keeping back straight, head up and a fairly fast drive up with the weight. Go down far enough so that the thighs are parallel with the ground. Squats can be performed flat-footed, or with the heels on a 2 or 3-inch (5 cm to 8 cm) block, or you can go up on to the toes each time to get really good results with a double effect. Jump squats with dumbbells held at sides. Approximately starting weights 20, 25, 30 lb a piece. Three sets of 15 reps. Two-arm dumbbell prone press on bench. Approx. starting weight 15, 20, 25 lb a piece. Three sets of 8 to 10 reps. Concentrate on a fast type of press. Sit ups. Start[...]

the heart

he has to be tactfully blunt

The best way to assimilate calcium is with vitamin D as a catalyst and preferably in a gluconate form for easy absorption. I always get my athletes to use Calcium Sandoz before marathons or triathlons.

Ask people who are successful coaches or have been in the sport for a long time. They have vast reservoirs of knowledge, which they may not realise, but tap them and you'll make your own progress that much easier because you'll know the right things to do and the wrong things to avoid.

High knees Keeping the upper body as relaxed as possible, move the legs as fast as you can, with a high knee-lifting action. The forward momentum should be slow. Keep up on your toes throughout the exercise. Straight back legs. Bounding By running with a bounding action and a good knee lift, concentrate on working the ankles hard. Drive hard with each stride. Up a slope is more effective. Running tall Concentrate on keeping high up on the toes and lifting the body upwards while you're running. Now combine all three of these actions while running fast. High knees, using your ankles for driving forward and running tall. Cool down with a 15-minute jog.

Indifference is a problem. An athlete can show little or no reaction to poor personal efforts and lack application.

drop the long repetitions in favour of short ones and wind sprints to sharpen the whole body and its reactions

If we're going to develop muscle bulk through the red fibres, we can use weights and resistance exercises, progressively increasing the periods of exercise and the weights and resistance employed.

chromium expert, the late Richard Doisy, recommended a supplement of two tablespoons a day.

The training must be varied so the athlete is not doing the same thing over and over again.

Cleaving strictly to a given schedule is not the way to progress.

Avoid fatty foods; they slow paristalsis and, therefore gastric emptying. Eat two servings daily of fresh fruit (one to be citrus fruits or tomatoes). Eat vegetables daily, including leafy, green vegetables (salads) and roots and tubers (turnips, beet, potatoes, etc.). Eat at least three slices of whole-grain bread daily. Eat enough butter or fortified margarine to supplement the bread. Drink at least three glasses of milk daily. Watch yor salt intake. In small doses, it is immediately eliminated from the body through sweat and urine; in large doses, it is contained in the body tissues and blood stream, resulting in a state of hyperchloremia.

there was the best training he could get. None of the American way: The guy with 'Coach' on his back, a clipboard and a stopwatch in his hands, shoving kids through repetitions until they are falling down with fatigue, blacking out and vomiting because the oxygen debts they are incurring are so great their central nervous systems are being attacked. Many of them, who got their college education only because they won sporting scholarships, have told me since that they dropped running as soon as they graduated although they slowly drifted back to jogging and social running with friends and family. They had all had the competitive edge drilled out of them.

cottage cheese, Swiss cheese, yoghurt

try all the shoes available for lightness, then find out which ones bend easily at the arch – in the middle – both ways, upwards and downwards. Look for the shoes that don't have too much on the uppers. You don't want stabilising bars and all this paraphernalia; the less you have on the top the better.

when you begin running, the only thing you should be thinking of is the development of muscular endurance, of building capillary beds, of regular, long, easy training periods, at least two hours or more if possible. Don't stop for more than 12 or 15 seconds.

It is better to prevent than it is to cure.

In shorter races, say, under 10 km, runners burn glycogen almost exclusively

B2 (riboflavin) helps the digestion of fat

it can be psychologically uplifting to know that you're out there doing it when other people are huddled inside

we must have strong upper leg muscles with full endurance.

Increases in the weight when you're performing three sets could be, for example, 5 or 10 lb (2 or 4 kg) from the first set to the second and another 5 or 10 lb from the second to the third. This would give you, for example, sets at 90, 95 and 100 or at 90, 100 and 110. According to reactions and progress, weights can be increased each week by 5 or 10 lb.

quadriceps

Not all of us can run sub-four miles or sub-thirty ten kilometres – but we can find distances and events at which we can enjoy degrees of success, even if it's only personal satisfaction.

standing on a stair on the balls of the feet and lifting up and down to give full extension to the sinews and muscles both in front of and behind the legs

even if he feels good on the way back, he doesn't make the mistake of finishing with a sprint. The ideal way to finish is always to feel that you could run some more.

mere repetitions of the same sub-maximal effort cause no change in muscle strength.

they should analyse the performance objectively and precisely and learn from the experience, rather than hurling their running shoes around the dressing room and berating themselves. Every error or failure should be a learning experience.

development racing – under-distance and over-distance events – at least once a week with another race adjusted according to your reactions in the previous race and the mood you're in.

They could dominate if they stopped applying so much pressure on their athletes from school age onwards.

Selenium

The coach, if he wants to get the best out of them, has to try to understand their differing points of view, any problems they may be facing at home and at work or school.

Your condition changes, the conditions outside change and that affects you and the times you're doing. Learning to make allowances is part of the mental approach to training efficiently.

Eliminate as much as possible foods which furnish only calories without contributing their share of vitamins and minerals (candy, cakes, carbonated beverages, etc.) and substitute fruit and fruit juices for desserts and snacks.

develop or maintain that level as they grew older. But the boys they had been beating, who didn't have the natural talent but had worked harder and more sensibly at developing their running, and maintained a high oxygen uptake capability as a consequence, went on to be the champions.

If you toss and turn, turn to magnesium

The shoe for you should not have any pressure points at all. Close your eyes and concentrate; if you feel any pressure anywhere, that's where the shoe is likely to give you trouble, so it's not for you.

you 've got to get your muscular system into condition so you can add power and speed later without muscle pulls or other damage. That means a big initial mileage. You concentrate initially for several months on purely aerobic running. Fast aerobic running if possible, keeping the effort just below that point where it can overbalance into anaerobic running. You must do as much of this aerobic running as you can. The minimum is three months. Four months is better, five is better still but anything less than three months is not enough.

During aerobic exercise, the body uses for energy about 48 per cent carbohydrates, 48 per cent fat acids and 4 per cent protein. In anaerobic exercise, the ratio changes to about 60 per cent carbohydrates; 25 per cent fat acids and 15 per cent protein

red and white fibres (otherwise known as slow twitch and fast twitch), you have to define exercises which work on both

B3 (niacin). The most important involves the red blood cells, which carry oxygen to all parts of the body

grain breads and starchy vegetables and the natural sugar source of honey

We've got to be able to straighten the back leg and drive

Shin splints are membrane ruptures between the muscle and the bone and often arise because of the jarring of downhill running or overstriding. You can counter overstriding by building up the front of your shoes a little and you should always be careful when running down hills. Water therapy, cold packs and heat treatment can help this injury.

the activity that best produces this consistent aerobic pressure is cross-country skiing

The immune system must be strong enough to resist infections when the whole body is under the stress of a tough training regime

The best thing is to do all your conditioning at sea level and then go up. You can do speed work and so on during the six to twelve weeks needed for your normal blood change.

The cold brings the natural cortisone to the area, which stimulates the circulation and helps

It is found that the maximum increase in muscle strength is obtained with one training stimulus a day.

Let's consider the qualities of the top athletes:

your energy must be conserved for the races.

I recommend that an aerobic run over five kilometres or a test of how far you can run aerobically in fifteen minutes are the best for checking fitness progress.

Any athlete sent out to do something taxing without being given a reason for it is unlikely to put his heart and soul into it

The coach must develop a positive attitude in his athletes.

Hill running also helps flexibility.

When we are training for athletics, we don't introduce the specialty exercises until we get into the last 10 weeks of a training programme. Until then, the 800-metre runner does much the same conditioning work as the marathoner.

enjoying running as a pleasant exercise within their limitations. Setting them out to beat the other kids is contrary to the development of future champions.

Good sources are whole grains, soybeans, nuts, green, leafy vegetables, fruits and blackstrap molasses but best of all is dolomite, a preparation of powdered dolomite limestone, which delivers magnesium and calcium in the exact proportions nature intended.

it is not the best athletes who succeed in important competitions but the best prepared.

The athlete must understand the training schedule he is following, its mechanical and physiological effects and the reasons why its pattern needs to be followed as closely as possible, day by day, as a systematic method of developing full potential

In the two days before competition, when training should be light, the taking of extra glucose or fructose can help in building up the body's glycogen content.

improvement in both muscular and mental endurance.

Folate, best found in green, leafy vegetables, particularly from the cabbage family, but also in wholewheat, brewers's yeast, oranges, beets, beans, meat and eggs, helps to form genes, the chemicals that blueprint every cell, including the oxygen-carrying red blood cells. Folate deficiency equals anaemia.

If a supplement is needed, I recommend honey. It's easy to digest, it has high calory value, it's not bulky and it doesn't cause gases. We found that, using honey as the supplement, we could eat perfectly normal meals and still have sufficient energy. Liquid glucose is another good substitute.

The ankles can develop more speed than a lot of people realise

The back of the shoe should not bite on the tendons and the joint on the inside of the foot should not be touching the shoe.

To improve our ability to transport more oxygen and then use it and our blood sugars through muscular endurance, we need good capillary beds. These can be developed quite substantially by continuous aerobic use of muscle groups over long periods.

Taken daily, it will prevent cold. It's a natural antihistamine, so stops hay fever. And it beats the heat. Increasing your C intake a week or so before an event in hot weather will increase your competitive edge.

Do we need specific reasons to run? That means motivation. We all need that, whether we intend just to get ourselves into a reasonable condition, or go out with friends who are social sportspeople or aspire to be great athletes. At all levels, we need to have our goals.

The hamstrings should be kept stronger than the quads. Cycling or running up stairs and hills will eliminate that problem

When training sessions are held only every second day, the increase in strength is about 80 per cent; with two training sessions a week, the increase is about 60 per cent; and when training sessions are held only once a week, the increase is only about 40 per cent of the improvement in strength gained through the daily regime. One training stimulus every 14 days produces no change at all in muscle strength.

what this book is all about: Guiding you to get all those 21 elements together in the right place at the right time for you to attain your selected goals in athletics.

qualities in the top coaches

The hard week brings in some harder anaerobic volume work, some fartlek, some long aerobic running, some technique or actual speed training and probably a couple of trials run at less than full effort to bring about co-ordination. You do your longish run as well.

Dr Uhlenbruck came to the conclusion that if you use muscle groups continuously for long periods, even at the low levels adopted by some ultra runners in training, you very quickly develop the dormant capillary beds and also establish new ones and, as well, a mitochondria that will be likely to remain indefinitely

Have a team talk on tactics two nights before the game and don't mention the subject again.

if they let me down once, I never let them put it over me again.

transitory phase of about four weeks

Take these two words with you into your competitive season: fresh and sharp. Always go into a game fresh and sharp so never do training during the week which is too exacting. You have trained to compete, not to keep on training.

Girls can run just as easily as boys; when they're young, they can often run better.

Athletes who do not sweat do not need extra potassium. Those who do and take salt tablets need double doses

Elation is natural, through success, through personal improvement, even if the race hasn't been won, because they vindicated the faith of others or even just because they took part in an event. It's not an emotion to be squashed but encouraged.

Placing trust in your coach is essential but your own input – through reading all you can about your particular sporting area, by learning at least the basics of physiology and so on – plays an important part.

It is unwise to eat sugar within three hours of a game.

The high oxygen uptake levels of properly trained athletes feed a better oxygen supply to all parts of their bodies, including their brains.

Fresh oranges are rich in vitamin C, as are all citrus fruits, and, surprisingly, so are potatoes, green peppers, parsley

How to run well has to be taught and practised

We can improve speed with better technique but we cannot basically make a slow person very fast. But when it comes to endurance, we can develop that in anyone.

Watch shoe wear closely because excessive wear leads to stress right through the leg to the hip and to bone wear.

Adenosine triphosphate (ATP) is the chemical which is the source of our energy.

it would be wise to find out little more about the quality of the soils from which your food comes.

Running will improve your mental capacity and confidence

Running, for instance, entails getting the right stride technique and then adding a longer and faster stride to produce the results.

Muscular endurance is the quality which enables muscles to maintain contractions in conditions of fatigue

What we need to aim for is both fitness and health

understanding of the physiology of athletics without trying to be an exercise physiologist. An understanding of psychological fundamentals without trying to be a psychologist. An understanding of mechanical fundamentals without trying to be a mechanic

You can comfortably run a half-marathon once a month or every six weeks and you can run a marathon every couple of months, provided you make sure you get full recovery from hard racing

When you start jogging, you are almost certainly going to get sore muscles. By all means, try massage; but the important thing is to out and jog again the following day, even if it's for only a slow ten or fifteen minutes at the most. Allow your heart to push the blood around and raise the blood pressure and use the exercise to flush out waste products which are causing the muscular discomfort. Let the heart do the gentle massage for you. If you stop the exercise until the muscle ache disappears, you'll have to begin again and work through sore muscles again. You'll have gained nothing

Anaerobic running and exercise are necessary – but their value and use should be clearly understood before they can be used successfully.

Some you have to pat on the back, some you have to kick in the backside. You've got to make them tougher, make them realise there is no easy way, it hurts everyone.

three days a week of heavy overload work and in this you can do any kind of anaerobic work you feel like. You can run straight-out long distance hard, say, three, five or ten kilometres; you can do hard repetitions over whatever distances you feel like; you can run hard on forest trails, if such luxuries are available to you.

The warm-up should include suppling and stretching exercises to relax the muscles and stretch the sinews. Some short sharp windsprints are desirable. The team should be taken outside and pushed through two or three 30 to 50-metre windsprints and then left to run or jog quietly until the game is called.

women, again as a general rule, have more subcutaneous fat in the muscles than men and, when it comes to real endurance, they seem to have an advantage. It's a natural storehouse of energy which they can use before running into deficit.

eat natural unprocessed foods, which provide a natural amount of sodium and taste better

Athletes need to be told why specific aspects of training are being used, both physiologically and mechanically.

poorer coaches may get early results but they can't continue to develop their athletes' potential beyond that, usually because they have overdone the anaerobic training far too soon in the interest of achieving that early success.

Drinking alcohol within 12 hours of a game prevents the red cells from combining with oxygen with maximum efficiency, because the alcohol is absorbed by the red cells. Carbonated drinks have the same effect so you're wise if you avoid them as well.

They run with no pressure on them, no racing except in fun, and all the time they are laying a wonderful foundation of high oxygen uptake and endurance so that, when they finally turn out in a race somewhere, many of them produce fantastic times

Bioflavonoids are vitamin C's close cousins. The white inner skin of citrus fruits and the white column inside a green pepper are packed with them and their great value is that they strengthen capillaries.

Check your footfall by running on sand or across a dewy lawn. To be most effective, your feet should form almost a single straight line.

That's why my athletes always ran up hills and stairs – to get that balance right. I never had any problems with runners.

Running hills/stairs > weightlifting

Check your shoes every week. If you are running correctly, the main wear should be on the outside of the heel, the outside of the forepart and slightly on the toes.

anaerobic system will function temporarily even during a long run at a basically aerobic work-rate – when you tackle a tough hill or kick in a burst of speed, for instance

Vitamin A

we get into the habit of training because we feel so good after it and, if we don't train for a day, we get hit by a guilty conscience. We develop an abundance of mental and physical energy and we tend to protect that jealously and zealously by never missing training or hating ourselves when we do.

So, a foot coming through low to the ground gives a slow stride; a foot coming through high gives a fast stride.

In your first workout, you should perform only one set of each exercise; in the second workout, two sets; in the third, three sets.

the only way we are going to get that higher oxygen uptake level is by steadily exercising over long periods – and the only way we can do that, day after day, is by maintaining the exercise at an aerobic level, not an anaerobic level.

Those people can't get up and motivate. They can't explain the important details even if they know because they don't spring readily to mind. If they've got to rifle through their notes for answers, they won't motivate.

when you race every marathon, which means you will come in very, very tired, you should aim at no more than one every two months at the most.

Cycling is a good exercise for cardiac development and general fitness if running isn't your style. You can row and get excellent results because it employs a legs-arms, arms-legs action. Swimming is beneficial but it does lose some effect because your body weight is being supported against gravity in the water.

Champions are developed, not born, although some have greater natural ability than others

Some athletes have said they have experienced a euphoria while running, a spiritual sensation of oneness, a communion with nature, the grass, trees, flowers, wild life and even God. These effects will occur if you let them and wish them to be there, if you are relaxed and taught to understand meditative exercising

I have never used interval training

Get out there and win and enjoy the process.

desired distance; the sustained sprints, from 200 to 400 metres, are unwise

the only source of iron the body will absorb is vitamin C

The athletes with the best motivation are those who are well informed, understand their training because it makes sense to them and have confidence in their coach

Quitting then is what you have to do

I believe the best tactic is to build up your blood sugar level the day before you compete and eat sparingly on the day of competition, which rules food supplements out as a pregame meal.

they cannot stand heavy doses of anaerobic training and pressure from coaching regimes to race a lot.

extra calcium will unknot muscle cramps.

I'm talking simple exercises such as balancing on the balls of the feet on a stair tread or a step or a thickish book and raising the heels up and down; or putting a rope or cycle tube round the front of the foot and gradually pulling upward at it; or picking up objects with the toes and so on. These all strengthen the muscles of the feet. They cost nothing.

If you get sore muscles and stop training until they recover, you're going to have a continual problem.

Hamstring trouble is caused by leg-speed running if muscles and sinews aren't stretched

This should be borne in mind to guard against going out too fast at the start of a race or training session or failing to warm-up adequately to prepare the aerobic mechanism

too much fast food

you run twenty miles one day and only ten the next or you run fast one day and slow the next over more similar mileages.

Something has to go back when you're running; it should be your hand

As a general rule, aim to do eight to ten exercises in each workout. It is a good idea to alternate the exercises between upper-body and lower-body muscle groups to spread them evenly throughout the session.

aerobic metabolism is far more efficient than anaerobic metabolism because more energy is derived from a given amount of foodstuff when the reactions occur under aerobic rather than anaerobic conditions.

A top coach is an affable, easy-going personality. He has the ability to joke with his athletes so that the enjoyment of each other's company is mutual. He has to be understanding and sympathetic.

balance sit-ups with back-arching exercises

Your objective must be to make yourself a fitter person, not to beat the chap down the road.

Here is a typical sprint workout: Warm up for about 15 minutes by running easily for 7, doing two or three stride-outs over 60 metres, suppling and stretching with exercises until your body feels loose and then, with 3-minute intervals between runs, go twice through the following exercises over 100 metres:

We must appreciate that sometimes, for reasons beyond control of the athlete or the coach, something can happen and the athlete doesn't perform up to the hoped-for best. The

solution is to talk about it in a sensible way, analyse it and try to determine why the bid didn't succeed. This isn't looking for excuses but finding analytical reasons so that the same mistake isn't made again.

in the last six weeks, you get into co-ordination work, getting the body used to the speed and distances at which you plan to race.

what I have them doing is all aerobic and that's good for them. I do not use anaerobic training.

Studies have shown that people with normal blood levels of iron who added iron to their diets increased their work capacity four times, which should be significant for hard-training athletes. But be sparing. It is possible to overdose on iron. Vitamin C should be enough and maybe 20 mg of an iron supplement – but no more.

You have to hurt yourself a little but at the same time you must enjoy it.

Learning to quit when your body says it's had enough is important.

If you like it, eat it; if you don't, it's not really going to do you all that much good.

Kenyans had a higher concentration of the enzymes which break down fat and greater quantities of citrate synthase, a critical enzyme needed to provide muscles with energy aerobically

Protein forces calcium out of the body, another argument for taking supplements.

light, flexible mid-sole, plenty of resilient cushioning under the feet, allowing your foot to function as naturally as possible.

His secret is that he sets his bike in a high gear and heads up the hills. This gives him superb muscle balance.

Just get into a tepid pool with a kickboard – and kick.

When your muscles are metabolising mainly fat, the oxygen demand is greater than when burning glycogen, which means you either have to take in more air or slow down. Fat, therefore, is not the preferred fuel for fast running.

The most concentrated source is fish liver oil; and vitamin A supplements, virtually tasteless and odourless, are the best source of fish liver oil. A supplement of 10,000 IU a day is enough but you can take up to 30,000 a day without worry. And, if you take vitamin E with it, the body's ability to use vitamin A increases sixfold.

Let them train at their pace and their degrees of effort; you train at your pace and effort levels.

You've got to learn to relax and let the arms come through in a loose, relaxed manner.

Chin ups. Two or three sets of what you're capable of. Palms of hands facing away, repeat with palms facing towards as separate exercise. Finger flips off a wall, three sets of 10 to 15 reps. Use the fingers only and lock the arms. Finger push-ups off the floor, three sets of 10 to 15 reps, start on finger-tips and push up with back straight.

We have to teach that pump to work progressively harder to take more blood with more oxygen to where it's wanted.

coach looks at basic speed as the governing factor and steers them to where that basic will work best.

prevention is flexible ankles

A good coach, for instance, doesn't have injured athletes.

The maximum training effect is achieved by using only 40 to 50 per cent of the maximum strength in voluntary isometric muscle contractions.

Sometimes, we've got to make them realise that maybe it's because their background of training isn't sufficient, they haven't been at it long enough, they might lack the speed which means they're racing over the wrong distances. There are always ways to build up confidence but it needs to be approached carefully.

You mix it with easy fartlek – striding up and down hills, jogging, doing exactly what you feel like and enjoying it, which is what fartlek (speed play) is all about.

How you run should be governed by how you feel on the day and by the simple catch-phrase I invented years ago, “Train, don't strain.”

Oysters contain 100 times more zinc than any other food but other good sources are red meat, liver, wheat germ and nuts. Unless you pig out on oysters, a 20 mg a day supplement of zinc is recommended

Coaches need to be sensitive to the emotional factors which influence their athletes

The goals in my schedules are optimum guidelines; they have to be modified to suit the conditions on the day, how you feel and so on.

Distribute the consumption of food over three regularly-spaced meals. To gain weight or prevent weight loss, an evening snack can be added.

There have been many examples in New Zealand of top high schoolboys who, on natural ability, could beat everyone in the distance races, the road races and the cross-country events but then, at twenty or so, were no longer champions. With maturity, people lose the high oxygen uptake which is natural in children and these top runners hadn't trained to

Everyone already gets too much and it needs to be as near as possible to a one-to-one ration with calcium – so get into those calcium supplements and watch the meat intake because meat has a phosphorous to calcium ratio of about 209 to 9.

You should be able to have your toe close to the end of the shoe but the shoe should not pull down on it as you move.

Once you've built that foundation, you can increase the speed a little,

The wise course is to eat the foods you feel like eating, as long as they are fresh and wholesome, rather than making drastic changes to foods you might not enjoy as much. If you cannot get the wholesome food you need, then take multi-vitamin tablets, even though some of you may be sceptical of their value. Vitamin supplements, in my experience, can definitely be valuable.

Once you have established the base of muscular endurance, maintaining muscular strength and tone is attainable by developing the muscle fibre through the application of resistance – by lifting weights or running up stairs or hills.

milk and cheese, almonds and liver. Any whole grain will do but wild rice is best.

One benefit of training in this disciplined way is that you quickly find that discipline extending to other aspects of your life and, for most of us, that is a valuable bonus

flexible, powerful ankles.

Weight training could be done three days a week until about two weeks before the season begins. Then reduce to two sessions a week until you begin playing and then back to a session a week, early in the week, throughout the playing season.

Running lifts the body weight against gravity, largely using the upper leg and thigh muscles hard enough and long enough to get a better result even than cycling, which ranks next, rowing or walking.

Diet is another factor which comes within the scope of the good coach. He needs to know that his pupils are getting the right elements, such as zinc for recovery from anaerobic effort

you cannot train as fast and as far continually at altitude as you can at sea level.

Then go out to 45 minutes, with two 15-minute days in between, and then on to an hour plus two 15s when you can handle it. Then you can start to bring up the intermediate days – alternating an hour, two half-hours, an hour and so on.

Maintaining a maximum isometric contraction for one or two seconds is sufficient to provide a training stimulus. When the contraction involves only two-thirds of the maximum strength, it should be maintained for between four and six seconds and so on

Don't let the anxious athlete go to bed too early; he should stay up and play cards, go to a movie, be with people who won't even talk about running and then go to bed when he or she feels tired.

Once a week, you go for a long, steady run of an hour or an hour and a half. Once you get into racing, you train to race, you don't train to train, so the work is very light and anything anaerobic is short and sharp. Any long running is not too far or too fast.

You cannot classify anyone by age. The fit and the unfit are there in all age groups.

A must for normal bones and muscles and best found in beans, peas, spinach, bran, brussels sprout and blueberries.

Coaches should welcome questions regarding training and other methods.

Get out and do some easy jogging as a supplement to your scheduled training and mix in a little bouncing, springing and running up hills or flights of stairs to apply resistance to the muscle fibres and develop more power and strength.

Eliminate coffee, tea and alcohol, which usurp more nutritious foods and can cause undesirable pharmacological effects, such as decreased muscular efficiency.

if we are going to train children, we 've got to encourage them to see how far they can run, not how fast

Enough magnesium and muscles relax; too little and they twitch and tremble.

Watch heel wear on your shoes closely. Excessive wear leads to stress right through the leg and hip and can also cause bone wear.

Training is essentially an individual thing. No two athletes are exactly the same; they have different strengths and weaknesses and only the best coaches can evaluate those differences and set out a programme that will strengthen the good points and eliminate the weaknesses.

